

HAT-P-36b

R-0834

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250215 · created 2026-07-28T17:54:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460721.9139 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.03
Transit depth (Rp/Rs)^2	2.46 +/- 0.94 [%]
Orbital Inclination (inc)	89.21 +/- 2.67
Airmass coefficient 1 (a1)	0.034 +/- 0.013
Airmass coefficient 2 (a2)	2.99 +/- 0.19
Scatter in the residuals of the lightcurve fit is	33.09 %
Best Comparison Star	None
Optimal Aperture	43.29
Optimal Annulus	94.53
Transit Duration (day)	0.0767 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.157 $\pm$ 0.03 vs 0.126 (deviation 0.32 σ)
Duration fit vs published	0.0767 d vs 0.0925 d (deviation 0.87 σ)
Mid-time O-C	-15.0 min
Depth SNR	1.6
Residual scatter	33.09 %

Light curve

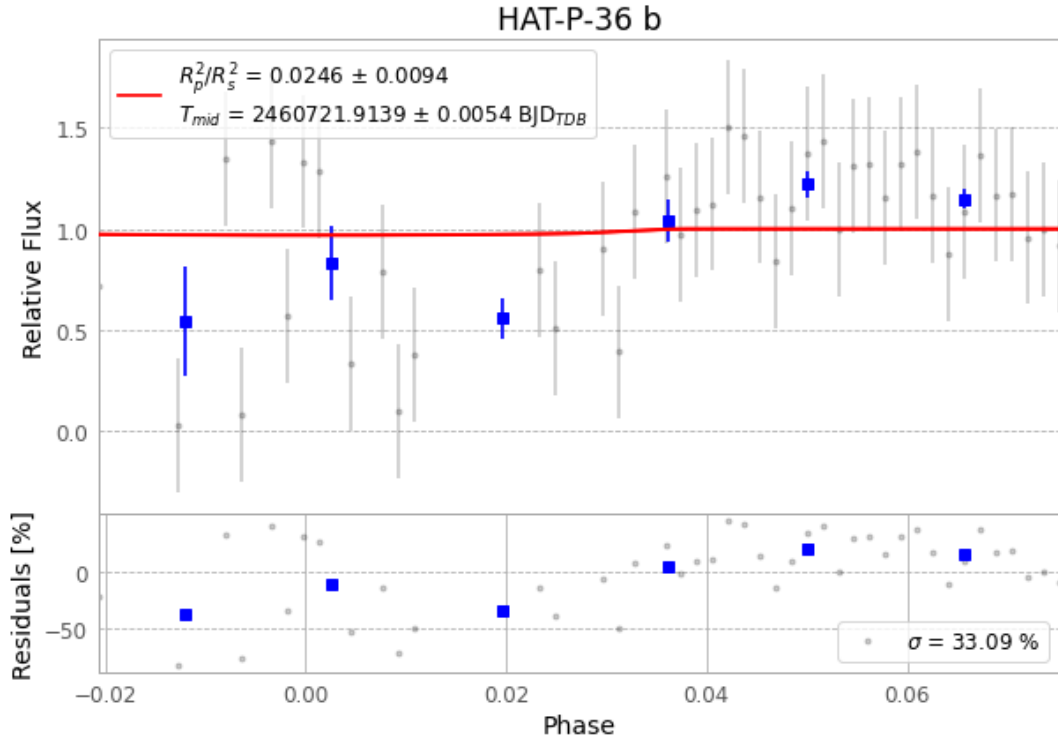


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-02-15.png (SHA-256 bc065c8d24504d88...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 366], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 60c87d029cb8d958...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

63 FITS frames (41 MB), HATP-36250215090615.FITS → HATP-36250215121215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250215.log (62a33836653d...), FinalLightCurve_HAT-P-36 b_2025-02-15.png (bc065c8d2450...), FinalLightCurve_HAT-P-36 b_2025-02-15.csv (af1c03560dd1...)

Compute resources

Wall time 556.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 17:54Z.